

Rebase vs merge

Two branches diverged, now you want them together

When to pick which

- Nobody else has built on your commits
- Merge when integrating into a shared branch
- Never rebase commits others are using

Browser lab

Digital Design and Verification Platform

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Tools

INTERACTIVE TOOL

Rebase vs merge chooser

Pick **rebase** or **merge** for chip-team scenarios — see the resulting history shape and why one choice fits (no interactive rebase).

Starter example: Private feature updating from main — prefer rebase for a linear PR. Shared branches prefer merge.

Load starter example

Challenges 0 / 22 cleared

Quiz: **rewrite**: Which rewrites commit SHAs? Answer: rebase

Answer

Show hint

Check

Next

Idle

Quiz: rewrite

Quiz: keep SHAs

Quiz: shared

Quiz: private

Pick: private

Pick: shared

Pick: land on main

Pick: solo remote

Pick: hotfix

Pick: cleanup

Pick: release

Pick: PR update

Quiz: force

Quiz: linear

See after graph

Quiz: no -i

Five correct

All eight

Quiz: M

Quiz: published

Fix shared pick

Starter scenario

Rule of thumb

Rebase when

Branch is yours alone (or team agrees). You want a linear story onto latest main.

Merge when

Commits are shared / published. You must not rewrite SHAs others already pulled.

Rebase vs merge chooser starter

learn_git — git rebase merge

Real Git practice

```
wsl - module11-git-rebase-merge/examples/rebase

# Track A - real Unix
# real shell session (Track A)
# practice repo: /tmp/git-rb-Y9j9L4

$ git init

$ echo "base" > notes.md

$ git add notes.md

$ git commit -m "Base"

$ git branch -M main

$ git checkout -b feature/update

$ echo "feature" >> notes.md

$ git add feature.txt

$ git commit -m "Feature work"

$ git checkout main

$ echo "main update" >> notes.md

$ git add notes.md

$ git commit -m "Main moved forward"

$ git checkout feature/update

$ git rebase main

$ git log --oneline --graph --all
* 7645b82 Feature work
* 2bc7997 Main moved forward
* 0d1bd70 Base
```

Real shell — rebase feature onto main

Real Git practice — try these

```
# git init — create a new repository here  
git init
```

```
# echo ... > notes.md — base commit on main  
echo "base" > notes.md
```

```
git add notes.md
```

```
git commit -m "Base"
```

```
git branch -M main
```

```
# git checkout -b feature/update — start feature work  
git checkout -b feature/update
```

```
echo "feature only" > feature.txt
```

Pitfalls to watch

- Do not rebase main or any branch teammates share without agreement
- After you rebase a branch you already pushed
- Merge is fine when you want to preserve the exact integration moment
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, load the starter and justify a few rebase versus merge picks
- On real Git, rebase a feature branch onto main and read the graph
- When you are ready, take the short quiz, then continue to the next module on cherry-pick